

# The modified exponential time differencing (ETD) method for solving the reactor point kinetics equations



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## ABSTRACT

The exponential time differencing (ETD) method with Taylor's series approximation is developed to solve the reactor point kinetics equations using large time-step. It is a semi-analytical and self-starting method in which the point kinetics equations are integrated using an integrating factor. The power and precursor concentrations occurring in the equations are expanded as a polynomial in derivatives. The coefficients of the polynomial are obtained from the recurrence relation. This method is applied to estimate the power transients of both thermal and fast reactors with multi group of delayed neutrons and the results are compared with other standard methods. From the results it is found that the modified exponential time differencing method estimates the power transients accurately for thermal and fast reactors and the numerical solution is found to converge fast. The error analysis of this method is discussed and a criterion for choosing the time-step for step and slowly varying reactivity insertions is also presented.

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## 1. Introduction

The power transients taking place in nuclear reactors depend on the kind of reactivity perturbation acting on the reactor and also the power level in which the reactor is operating. It is important to predict the power transients, caused by reactivity perturbations, during normal operation and accidental conditions. Though the space time kinetics represents the time-dependent behaviour of nuclear reactor in a more realistic way, solving the space time kinetics equations consumes more time. Determination of spatial variation of flux in a reactor is more difficult than the determination of power. So, generally in medium sized thermal reactors and tightly coupled fast reactors, the point kinetics is found to be satisfactory in predicting the power transients. The point kinetics equations with multi group of delayed neutrons are stiff non-linear differential equations. A major difficulty in numerically solving the point kinetics equations arises from the stiffness term, which necessitates the requirement of choosing very small time-step. There are different kinds of methodologies available in the literature to study the reactor transients. Hennart and Barrios (1976) applied the Pade and Chebyshev type approximation to solve the point kinetics equations. One can use the third order Hermite Polynomial Method (HPM) (Yeh, 1978) to solve the point kinetics

equations. Aboanber and Nahla (2002a,b) presented the analytical inversion method for solving the point reactor kinetics equations with multi group of delayed neutrons. Li et al. (2009) introduced a numerical integral method using the better basis function (BBF) and investigated the power transients caused by various types of reactivity insertion for thermal reactors with multi group of delayed neutrons. Recently Nahla (2011) applied the Taylor's series method (TSM) to solve the point kinetics equations.

In this work, the exponential time differencing (ETD) method with Taylor's series expansion is developed and presented for solving the point kinetics equations with multi group of delayed neutrons. The exponential time differencing (ETD) method was originally proposed in the field of computational electrodynamics (Taflave, 1995) to solve stiff differential equations.

This method is developed by integrating the point kinetics equations over a single time-step using an integrating factor. The power and precursor concentrations in the equations are expanded as a polynomial in derivatives and then integrated. The coefficients of the polynomial (ETD coefficients) are obtained from integration of simple function and a recurrence relation between them is also obtained. The recurrence relation is used to determine higher order ETD coefficients. This method is applied to estimate the power transients of thermal reactors and the results are compared with other standard methods referred by Nahla (2011) and Li et al. (2009). This method is also applied to estimate the power transients of the Indian Prototype Fast

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Breeder Reactor (PFBR) at IGCAR Kalpakkam and the results are compared with the [Cohen's method \(1958\)](#). From the comparison of results, it is learnt that the modified ETD method estimates the power transients accurately using large time-step for thermal and fast reactors with multi group of delayed neutrons. The results also show that the power transients may be estimated for longer duration of time. The error analysis of this method is discussed and the numerical solution is found to converge fast as the order of the ETD coefficients is increased. From the error analysis, a novel way of choosing the time-step, 'h' and its upper bound for convergence of the numerical solution are also derived and presented.

## 2. The modified exponential time differencing (ETD) method and the point kinetics equations

The exponential time differencing (ETD) methods are time integration techniques that provide accurate smooth solutions for stiff differential equations.

Consider the following stiff differential equation

$$\frac{du}{dt} = cu + F(u, t) \quad (1)$$

In the above equation 'c' is the stiffness constant which may be large and  $F(u, t)$  may be a nonlinear term. To numerically solve this kind of stiff differential equation, one should be able to handle the stiffness constant 'c', by properly choosing the time-step 'h'. To solve Eq. (1) using the exponential time differencing method, it is multiplied by an integrating factor  $e^{-ct}$  and integrated over a single time-step, from  $t = t_n$  to  $t = t_{n+1} = t_n + h$ , to get

$$\int_{t_n}^{t_{n+1}} \frac{d}{dt} (ue^{-ct}) dt = \int_{t_n}^{t_{n+1}} F(u, t) e^{-ct} dt \quad (2)$$

After integrating, we get

$$u(t_{n+1}) = u(t_n)e^{ch} + e^{ch} \int_0^h e^{-c\tau} F(u(t_n + \tau), t_n + \tau) d\tau \quad (3)$$

where  $\tau = t - t_n$ . The purpose of transforming the differential Eq. (1) into Eq. (2) is to remove the explicit dependence of the stiffness constant 'c' in Eq. (1). In the numerical solution, i.e., Eq. (3), the stiffness constant appears only in the exponential term. By removing the explicit dependence of the stiffness constant, the solution, Eq. (3), may be obtained using large time-step 'h'. Another major critical step in Eq. (3) lies in choosing a proper approximation for the integrand  $F(u, t)$  in the interval  $t_n < t < t_n + h$ . If  $F(u, t)$  is known a priori for  $t < t_n$ , then  $F(u, t)$  in the interval  $t_n < t < t_n + h$  may be obtained using interpolation with its previous values and the integral in Eq. (3) can be evaluated ([Beylkin et al., 1998](#); [Nie et al., 2006](#)). In case if  $F(u, t)$  is not known a priori, then  $F(u, t)$  in the time interval  $t_n < t < t_n + h$  can be obtained from the Taylor's series expansion of  $F(u, t)$ .

Consider the point kinetics equations given by

$$\frac{dp(t)}{dt} = \left( \frac{\rho(t) - \beta}{\Lambda} \right) p(t) + \sum_{i=1}^6 \lambda_i C_i(t) \quad (4)$$

$$\frac{dC_i}{dt} = \left( \frac{\beta_i}{\Lambda} \right) p(t) - \lambda_i C_i \quad (i = 1, 2, \dots, 6) \quad (5)$$

In the above Eqs. (4) and (5),  $p$  is the power,  $\Lambda$  is the prompt neutron generation time,  $\beta_i$  is the effective fraction of the  $i^{\text{th}}$  group of delayed neutrons,  $\beta$  is the total effective fraction of delayed neutrons ( $\beta = \sum_{i=1}^6 \beta_i$ ),  $\lambda_i$  and  $C_i$  are the decay constant and precursor concentration of the  $i^{\text{th}}$  group of delayed neutron. The initial

conditions of the point kinetics equations are chosen as  $p(t = 0) = p_0$ ,  $C_i(t = 0) = \frac{\beta_i}{\Lambda} p_0$ ,  $p_0$  is the steady state power before the introduction of any external reactivity. The neutron generation time is of the order of  $\sim 10^{-5}$  s for thermal reactors and it is of the order of  $\sim 10^{-7}$  s for fast reactors. Here the stiffness constant appears as  $\frac{1}{\Lambda}$ . To solve the point kinetics equations by the modified ETD method, Eqs. (4) and (5) are multiplied by the integrating factor  $e^{\frac{t}{\Lambda}}$  and integrated over a single time-step from  $t = t_n$  to  $t = t_{n+1} = t_n + h$ , to get

$$\int_{t_n}^{t_{n+1}} \frac{d}{dt} (pe^{\frac{t}{\Lambda}}) dt = \int_{t_n}^{t_{n+1}} \left( \frac{\rho(t) - \beta}{\Lambda} \right) pe^{\frac{t}{\Lambda}} dt + \int_{t_n}^{t_{n+1}} \sum_{i=1}^6 \lambda_i C_i(t) e^{\frac{t}{\Lambda}} dt + \frac{1}{\Lambda} \int_{t_n}^{t_{n+1}} pe^{\frac{t}{\Lambda}} dt \quad (6)$$

$$\int_{t_n}^{t_{n+1}} \frac{d}{dt} (C_i e^{\frac{t}{\Lambda}}) dt = \int_{t_n}^{t_{n+1}} \frac{\beta_i}{\Lambda} pe^{\frac{t}{\Lambda}} dt + \left( \frac{1}{\Lambda} - \lambda_i \right) \int_{t_n}^{t_{n+1}} C_i(t) e^{\frac{t}{\Lambda}} dt \quad (7)$$

After integrating and rearranging Eqs. (6) and (7), we get

$$p(t_{n+1}) = p(t_n)e^{\frac{h}{\Lambda}} + \int_0^h e^{\frac{t_n+\tau}{\Lambda}} \frac{\rho(t_n+\tau)}{\Lambda} p(t_n+\tau) d\tau + \left( \frac{1-\beta}{\Lambda} \right) \int_0^h e^{\frac{t_n+\tau}{\Lambda}} p(t_n+\tau) d\tau + \int_0^h e^{\frac{t_n+\tau}{\Lambda}} \sum_{i=1}^6 \lambda_i C_i(t_n+\tau) d\tau \quad (8)$$

$$C_i(t_{n+1}) = C_i(t_n)e^{\frac{h}{\Lambda}} + \frac{\beta_i}{\Lambda} \int_0^h e^{\frac{t_n+\tau}{\Lambda}} p(t_n+\tau) d\tau + \left( \frac{1}{\Lambda} - \lambda_i \right) \int_0^h e^{\frac{t_n+\tau}{\Lambda}} C_i(t_n+\tau) d\tau \quad (9)$$

Here, in the above Eqs. (8) and (9), we expand  $p(t_n + \tau)$  and  $C_i(t_n + \tau)$  as a polynomial in derivatives using Taylor's series expansion of order 'S - 1'. They are expanded as

$$p(t_n + \tau) = \sum_{k=0}^{S-1} \nabla^k p(t_n) \frac{\tau^k}{k!} \quad (10)$$

$$C_i(t_n + \tau) = \sum_{k=0}^{S-1} \nabla^k C_i(t_n) \frac{\tau^k}{k!} \quad (11)$$

where  $\nabla^k = \frac{d^k}{dt^k}$ . The reason for expanding the power and precursor concentrations using Taylor's series is that they require only present values and previous power history is not required. The local truncation errors in the Taylor's series expansion of  $p(t_n + \tau)$  and  $C_i(t_n + \tau)$  are termed as  $R_1(p)$  and  $R_2(C_i)$  respectively and they are given as

$$R_1(p) = \sum_{k=S}^{\infty} \nabla^k p(t_n) \frac{\tau^k}{k!} \quad (12)$$

$$R_2(C_i) = \sum_{k=S}^{\infty} \nabla^k C_i(t_n) \frac{\tau^k}{k!} \quad (13)$$

Substituting Eqs. (10) and (11) into Eqs. (8) and (9), we get

$$p(t_{n+1}) = p(t_n)e^{\frac{h}{\Lambda}} + \frac{1}{\Lambda} \int_0^h e^{\frac{t_n+\tau}{\Lambda}} \rho(t_n+\tau) \left( \sum_{k=0}^{S-1} \nabla^k p(t_n) \frac{\tau^k}{k!} \right) d\tau + \left( \frac{1-\beta}{\Lambda} \right) \int_0^h e^{\frac{t_n+\tau}{\Lambda}} \left( \sum_{k=0}^{S-1} \nabla^k p(t_n) \frac{\tau^k}{k!} \right) d\tau + \sum_{i=1}^6 \lambda_i \int_0^h e^{\frac{t_n+\tau}{\Lambda}} \left( \sum_{k=0}^{S-1} \nabla^k C_i(t_n) \frac{\tau^k}{k!} \right) d\tau \quad (14)$$

$$C_i(t_{n+1}) = C_i(t_n)e^{\frac{-h}{\Lambda}} + \frac{\beta_i}{\Lambda} \int_0^h e^{\frac{\tau-h}{\Lambda}} \left( \sum_{k=0}^{S-1} \nabla^k p(t_n) \frac{\tau^k}{k!} \right) d\tau \\ + \left( \frac{1}{\Lambda} - \lambda_i \right) \int_0^h e^{\frac{\tau-h}{\Lambda}} \left( \sum_{k=0}^{S-1} \nabla^k C_i(t_n) \frac{\tau^k}{k!} \right) d\tau \quad (15)$$

After interchanging the order of summation and integration in Eqs. (14) and (15), we get

$$p(t_{n+1}) = p(t_n)e^{\frac{-h}{\Lambda}} + \frac{1}{\Lambda} \sum_{k=0}^{S-1} \nabla^k p(t_n) q(k) \\ + \left( \frac{1-\beta}{\Lambda} \right) \sum_{k=0}^{S-1} \nabla^k p(t_n) u(k) + \sum_{i=1}^6 \lambda_i \sum_{k=0}^{S-1} \nabla^k C_i(t_n) u(k) \quad (16)$$

$$C_i(t_{n+1}) = C_i(t_n)e^{\frac{-h}{\Lambda}} + \frac{\beta_i}{\Lambda} \sum_{k=0}^{S-1} \nabla^k p(t_n) u(k) \\ + \left( \frac{1}{\Lambda} - \lambda_i \right) \sum_{k=0}^{S-1} \nabla^k C_i(t_n) u(k) \quad (17)$$

where  $q(k)$  is given as

$$q(k) = \int_0^h e^{\frac{\tau-h}{\Lambda}} \rho(t_n + \tau) \frac{\tau^k}{k!} d\tau \quad (18)$$

and the ETD coefficient  $u(k)$  of order ' $k$ ' is given as

$$u(k) = \int_0^h e^{\frac{\tau-h}{\Lambda}} \frac{\tau^k}{k!} d\tau \quad (19)$$

Eqs. (16) and (17) provide the numerical solution for the power transient and precursor concentrations using the Taylor's series expansion of  $p(t_{n+1})$  and  $C_i(t_{n+1})$  up to order ' $S-1$ '.

Substituting  $k=0$  in Eq. (19) we get

$$u(0) = (1 - e^{\frac{-h}{\Lambda}}) \Lambda \quad (20)$$

The recurrence relation of the ETD coefficient  $u(k)$  is given as

$$u(k) = \frac{\Lambda h^k}{k!} - \Lambda u(k-1), \quad k \geq 1 \quad (21)$$

### 3. Error analysis of the modified ETD method and control of time-step ' $h$ ' for step and slowly varying reactivity insertions

The final global errors involved in the numerical solution of the power transient (Eq. (16)) and the precursor concentrations (Eq. (17)) are determined as follows. By increasing the order of the Taylor's series expansion of  $p(t_n + \tau)$  and  $C_i(t_n + \tau)$ , Eqs. (16) and (17) are rewritten as

$$p(t_{n+1}) = p(t_n)e^{\frac{-h}{\Lambda}} + \frac{1}{\Lambda} \sum_{k=0}^{S-1} \nabla^k p(t_n) q(k) \\ + \left( \frac{1-\beta}{\Lambda} \right) \sum_{k=0}^{S-1} \nabla^k p(t_n) u(k) + \sum_{i=1}^6 \lambda_i \sum_{k=0}^{S-1} \nabla^k C_i(t_n) u(k) \\ + \frac{1}{\Lambda} \sum_{k=S}^{\infty} \nabla^k p(t_n) q(k) + \left( \frac{1-\beta}{\Lambda} \right) \sum_{k=S}^{\infty} \nabla^k p(t_n) u(k) \\ + \sum_{i=1}^6 \lambda_i \sum_{k=S}^{\infty} \nabla^k C_i(t_n) u(k) \quad (22)$$

$$C_i(t_{n+1}) = C_i(t_n)e^{\frac{-h}{\Lambda}} + \frac{\beta_i}{\Lambda} \sum_{k=0}^{S-1} \nabla^k p(t_n) u(k) + \left( \frac{1}{\Lambda} - \lambda_i \right) \sum_{k=0}^{S-1} \nabla^k C_i(t_n) u(k) \\ + \frac{\beta_i}{\Lambda} \sum_{k=S}^{\infty} \nabla^k p(t_n) u(k) + \left( \frac{1}{\Lambda} - \lambda_i \right) \sum_{k=S}^{\infty} \nabla^k C_i(t_n) u(k) \quad (23)$$

The final global errors  $\varepsilon_1(p)$  and  $\varepsilon_2(C_i)$ , involved in the computation of the power transient and precursor concentrations, (Eqs. (22) and (23)) are given as

$$\varepsilon_1(p) = \frac{1}{\Lambda} \sum_{k=S}^{\infty} \nabla^k p(t_n) q(k) + \left( \frac{1-\beta}{\Lambda} \right) \sum_{k=S}^{\infty} \nabla^k p(t_n) u(k) \\ + \sum_{i=1}^6 \lambda_i \sum_{k=S}^{\infty} \nabla^k C_i(t_n) u(k) \quad (24)$$

$$\varepsilon_2(C_i) = \frac{\beta_i}{\Lambda} \sum_{k=S}^{\infty} \nabla^k p(t_n) u(k) + \left( \frac{1}{\Lambda} - \lambda_i \right) \sum_{k=S}^{\infty} \nabla^k C_i(t_n) u(k) \quad (25)$$

For step (constant) and slowly varying reactivity insertions, the numerical solutions of the power transient and precursor concentrations i.e., Eqs. (22) and (23) are further simplified as follows. In this case  $q(k) \sim \rho(t_n)u(k)$ ,  $u(0) \sim \Lambda$  and  $u(k) \sim \frac{\Lambda h^k}{k!}$  for  $k \geq 1$  (since  $(\frac{1}{\Lambda}) \gg 1$  and  $\Lambda^{k+1} \ll \frac{\Lambda h^k}{k!}$ ). Substituting  $q(k)$  and  $u(k)$  into Eqs. (22)–(25), we get

$$p(t_{n+1}) = p(t_n)e^{\frac{-h}{\Lambda}} + (1 - \beta + \rho(t_n)) \sum_{k=0}^{S-1} \nabla^k p(t_n) \frac{h^k}{k!} \\ + \Lambda \sum_{i=1}^6 \lambda_i \sum_{k=0}^{S-1} \nabla^k C_i(t_n) \frac{h^k}{k!} + \varepsilon_1(p) \quad (26)$$

$$C_i(t_{n+1}) = C_i(t_n)e^{\frac{-h}{\Lambda}} + \beta_i \sum_{k=0}^{S-1} \nabla^k p(t_n) \frac{h^k}{k!} \\ + (1 - \Lambda \lambda_i) \sum_{k=0}^{S-1} \nabla^k C_i(t_n) \frac{h^k}{k!} + \varepsilon_2(C_i) \quad (27)$$

where  $\varepsilon_1(p)$  and  $\varepsilon_2(C_i)$  are given as

$$\varepsilon_1(p) = (1 - \beta + \rho(t_n)) \sum_{k=S}^{\infty} \nabla^k p(t_n) \frac{h^k}{k!} + \Lambda \sum_{i=1}^6 \lambda_i \sum_{k=S}^{\infty} \nabla^k C_i(t_n) \frac{h^k}{k!} \quad (28)$$

$$\varepsilon_2(C_i) = \beta_i \sum_{k=S}^{\infty} \nabla^k p(t_n) \frac{h^k}{k!} + (1 - \Lambda \lambda_i) \sum_{k=S}^{\infty} \nabla^k C_i(t_n) \frac{h^k}{k!} \quad (29)$$

Eqs. (26) and (27) represent the numerical solution for the power transient and precursor concentrations for step and slowly varying reactivity insertions. Eqs. (28) and (29) represent the corresponding global errors involved in the computation of power transient and precursor concentrations. The numerical solution of the power transient  $p(t_{n+1})$  and precursor concentrations  $C_i(t_{n+1})$ , (Eqs. (26) and (27)), converge if the global errors  $\varepsilon_1(p)$  and  $\varepsilon_2(C_i)$ , (Eqs. (28) and (29)), become small as the order of the ETD coefficients is increased, i.e.,  $\lim_{k \rightarrow \infty} \nabla^k p(t_n) \frac{h^k}{k!} \rightarrow 0$  and  $\lim_{k \rightarrow \infty} \nabla^k C_i(t_n) \frac{h^k}{k!} \rightarrow 0$ . At a first glance, the global errors  $\varepsilon_1(p)$  and  $\varepsilon_2(C_i)$  can be made to approach zero by choosing  $h < 1$ . But here the time-step  $h$  and its upper bound are mathematically derived as follows. For step and slowly varying reactivity insertion,  $\nabla^k p(t_n)$  and  $\nabla^k C_i(t_n)$  can be written in matrix form as

$$\begin{bmatrix} \nabla^k p(t_n) \\ \nabla^k C_1(t_n) \\ \nabla^k C_2(t_n) \\ \nabla^k C_3(t_n) \\ \nabla^k C_4(t_n) \\ \nabla^k C_5(t_n) \\ \nabla^k C_6(t_n) \end{bmatrix} = \begin{bmatrix} \frac{\rho-\beta}{\Lambda} & \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & \lambda_6 \\ \frac{\beta_1}{\Lambda} & -\lambda_1 & 0 & 0 & 0 & 0 & 0 \\ \frac{\beta_2}{\Lambda} & 0 & -\lambda_2 & 0 & 0 & 0 & 0 \\ \frac{\beta_3}{\Lambda} & 0 & 0 & -\lambda_3 & 0 & 0 & 0 \\ \frac{\beta_4}{\Lambda} & 0 & 0 & 0 & -\lambda_4 & 0 & 0 \\ \frac{\beta_5}{\Lambda} & 0 & 0 & 0 & 0 & -\lambda_5 & 0 \\ \frac{\beta_6}{\Lambda} & 0 & 0 & 0 & 0 & 0 & -\lambda_6 \end{bmatrix} \begin{bmatrix} \nabla^{k-1} p(t_n) \\ \nabla^{k-1} C_1(t_n) \\ \nabla^{k-1} C_2(t_n) \\ \nabla^{k-1} C_3(t_n) \\ \nabla^{k-1} C_4(t_n) \\ \nabla^{k-1} C_5(t_n) \\ \nabla^{k-1} C_6(t_n) \end{bmatrix} \quad (30)$$

By making use of Eqs. (4) and (5), Eq. (30) is rearranged as

$$\begin{bmatrix} \nabla^k p(t_n) \\ \nabla^k C_1(t_n) \\ \nabla^k C_2(t_n) \\ \nabla^k C_3(t_n) \\ \nabla^k C_4(t_n) \\ \nabla^k C_5(t_n) \\ \nabla^k C_6(t_n) \end{bmatrix} = \begin{pmatrix} \frac{\rho-\beta}{\Lambda} & \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & \lambda_6 \\ \frac{\beta_1}{\Lambda} & -\lambda_1 & 0 & 0 & 0 & 0 & 0 \\ \frac{\beta_2}{\Lambda} & 0 & -\lambda_2 & 0 & 0 & 0 & 0 \\ \frac{\beta_3}{\Lambda} & 0 & 0 & -\lambda_3 & 0 & 0 & 0 \\ \frac{\beta_4}{\Lambda} & 0 & 0 & 0 & -\lambda_4 & 0 & 0 \\ \frac{\beta_5}{\Lambda} & 0 & 0 & 0 & 0 & -\lambda_5 & 0 \\ \frac{\beta_6}{\Lambda} & 0 & 0 & 0 & 0 & 0 & -\lambda_6 \end{pmatrix}^k \begin{bmatrix} p(t_0) \\ C_1(t_0) \\ C_2(t_0) \\ C_3(t_0) \\ C_4(t_0) \\ C_5(t_0) \\ C_6(t_0) \end{bmatrix} \quad (31)$$

and

$$\begin{bmatrix} \nabla^k p(t_n) \\ \nabla^k C_1(t_n) \\ \nabla^k C_2(t_n) \\ \nabla^k C_3(t_n) \\ \nabla^k C_4(t_n) \\ \nabla^k C_5(t_n) \\ \nabla^k C_6(t_n) \end{bmatrix} \frac{h^k}{k!} = \begin{pmatrix} \frac{\rho-\beta}{\Lambda} & \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & \lambda_6 \\ \frac{\beta_1}{\Lambda} & -\lambda_1 & 0 & 0 & 0 & 0 & 0 \\ \frac{\beta_2}{\Lambda} & 0 & -\lambda_2 & 0 & 0 & 0 & 0 \\ \frac{\beta_3}{\Lambda} & 0 & 0 & -\lambda_3 & 0 & 0 & 0 \\ \frac{\beta_4}{\Lambda} & 0 & 0 & 0 & -\lambda_4 & 0 & 0 \\ \frac{\beta_5}{\Lambda} & 0 & 0 & 0 & 0 & -\lambda_5 & 0 \\ \frac{\beta_6}{\Lambda} & 0 & 0 & 0 & 0 & 0 & -\lambda_6 \end{pmatrix}^k \begin{bmatrix} p(t_0) \\ C_1(t_0) \\ C_2(t_0) \\ C_3(t_0) \\ C_4(t_0) \\ C_5(t_0) \\ C_6(t_0) \end{bmatrix} \frac{h^k}{k!} \quad (32)$$

Eq. (32) is rewritten in matrix form as

$$[Y^k(t_n)] = B^k[Y^0(0)], \quad (33)$$

where

$$[Y^k(t_n)] = [\nabla^k p(t_n) \nabla^k C_1(t_n) \nabla^k C_2(t_n) \nabla^k C_3(t_n) \nabla^k C_4(t_n) \nabla^k C_5(t_n) \nabla^k C_6(t_n)]^T \frac{h^k}{k!},$$

$$[Y^0(t_n)] = [p(t_0) C_1(t_0) C_2(t_0) C_3(t_0) C_4(t_0) C_5(t_0) C_6(t_0)]^T, \quad B^k = A^k \left( \frac{h}{k!} \right)^k$$

and

$$A = \begin{pmatrix} \frac{\rho-\beta}{\Lambda} & \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & \lambda_6 \\ \frac{\beta_1}{\Lambda} & -\lambda_1 & 0 & 0 & 0 & 0 & 0 \\ \frac{\beta_2}{\Lambda} & 0 & -\lambda_2 & 0 & 0 & 0 & 0 \\ \frac{\beta_3}{\Lambda} & 0 & 0 & -\lambda_3 & 0 & 0 & 0 \\ \frac{\beta_4}{\Lambda} & 0 & 0 & 0 & -\lambda_4 & 0 & 0 \\ \frac{\beta_5}{\Lambda} & 0 & 0 & 0 & 0 & -\lambda_5 & 0 \\ \frac{\beta_6}{\Lambda} & 0 & 0 & 0 & 0 & 0 & -\lambda_6 \end{pmatrix} \quad (34)$$

For convergence of the numerical solution for the power transient and precursor concentrations,  $[Y^k(t_n)] \rightarrow 0$  as  $k \rightarrow \infty$ . In other words  $\lim_{k \rightarrow \infty} [Y^k(t_n)] \rightarrow 0$  if  $\lim_{k \rightarrow \infty} B^k \rightarrow 0$ . This is possible if the magnitude of the largest eigenvalue of  $B < 1$ . Let  $\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5, \gamma_6, \gamma_7$  be the eigenvalues of  $A$  and let the magnitude of the largest eigenvalue of  $A$  be  $\gamma_{\max} = \max\{|\gamma_1|, |\gamma_2|, |\gamma_3|, |\gamma_4|, |\gamma_5|, |\gamma_6|, |\gamma_7|\}$ . Since  $B = A \frac{h}{(k!)^{1/k}}$ , the largest eigenvalue of  $B$  is  $\gamma_{\max} \frac{h}{(k!)^{1/k}}$ . Hence convergence is possible if  $\gamma_{\max} \frac{h}{(k!)^{1/k}} < 1$ . So the time-step  $h$  is chosen as

$$h < \left( \frac{1}{\gamma_{\max}} \right) (k!)^{1/k} \quad (35)$$

The upper bound for the time-step,  $h$ , is given as

$$\left( \frac{1}{\gamma_{\max}} \right) (k!)^{1/k} \quad (36)$$

where  $k > S - 1$ . With this time-step, the global errors  $\varepsilon_1(p)$  and  $\varepsilon_2(C_i) \rightarrow 0$  as  $k \rightarrow \infty$  i.e., both  $\nabla^k p(t_n) \frac{h^k}{k!}$  and  $\nabla^k C_i(t_n) \frac{h^k}{k!} \rightarrow 0$  as  $k \rightarrow \infty$ .

#### 4. Results and discussion

The modified exponential time differencing method is applied to solve the point kinetics equations with six groups of delayed neutrons for both thermal and fast reactors. The power transients for few types of reactivity perturbations i.e., step, ramp and oscillatory reactivity are computed and the results are discussed. The power transient initiated by a high reactivity insertion rate under unprotected accident scenario is also discussed. In all the power transient computations, discussed here by this method, the time-step is chosen to satisfy Eq. (35).

##### 4.1. Power transients of thermal and fast reactors caused by step reactivity

The modified ETD method is applied to estimate the power transients of the thermal reactor described by Nahla (2011). An external step reactivity  $\rho_0 = -1 \$$  is applied and the resulting power transient is analysed. The decay constants of the neutron precursors and the delayed neutron fractions for the thermal reactor are taken as  $\lambda_1 = 0.0127 s^{-1}$ ,  $\lambda_2 = 0.0317 s^{-1}$ ,  $\lambda_3 = 0.115 s^{-1}$ ,  $\lambda_4 = 0.311 s^{-1}$ ,  $\lambda_5 = 1.4 s^{-1}$ ,  $\lambda_6 = 3.87 s^{-1}$ ,  $\beta_1 = 0.000285$ ,  $\beta_2 = 0.0015975$ ,  $\beta_3 = 0.00141$ ,  $\beta_4 = 0.0030525$ ,  $\beta_5 = 0.00096$ ,  $\beta_6 = 0.000195$ , and  $\Lambda = 5.0 \times 10^{-4} s$ . Table 1 shows the values of the power transients obtained from this method using a time-step  $h = 0.1 s$ . Table 1 also shows the power transients obtained from the Taylor's series method (TSM) along with the exact values referred by Nahla (2011). The relative errors,  $|(X_{cal} - X_{exact})/X_{exact}|$ , are shown in brackets in Table 1. The comparison is made till 10s as the exact values are available till 10s only. From the comparison of results in Table 1, the accuracy of this method is assured.

Consider another example of thermal reactor described by Li et al. (2009) with the following parameters.  $\lambda_1 = 0.0127 s^{-1}$ ,  $\lambda_2 = 0.0317 s^{-1}$ ,  $\lambda_3 = 0.115 s^{-1}$ ,  $\lambda_4 = 0.311 s^{-1}$ ,  $\lambda_5 = 1.4 s^{-1}$ ,  $\lambda_6 = 3.87 s^{-1}$ ,  $\beta_1 = 0.000266$ ,  $\beta_2 = 0.001491$ ,  $\beta_3 = 0.001316$ ,  $\beta_4 = 0.002849$ ,  $\beta_5 = 0.000896$ ,  $\beta_6 = 0.000182$ , and  $\Lambda = 2.0 \times 10^{-5} s$ . A step reactivity  $\rho_0 = 0.003$  is inserted and the power transients are computed using the modified ETD method ( $h = 0.01 s, S = 7$ ). The results are compared with the Taylor polynomial method (TPM), the better basis function method (BBF) and the end floating method (EFM) referred by Li et al. (2009). The results are shown in Table 2. The relative errors are shown in brackets in Table 2. It is observed from Table 2 that the power transients computed by this method are more accurate than other standard methods as mentioned above. In this case also the comparison is made till 1 s as the exact values (Li et al., 2009) are available till 1 s only.

As an example of fast reactor, consider the Indian Prototype Fast Breeder Reactor (PFBR) at IGCAR, Kalpakkam. It is designed for 500 MWe. It contains 181 fuel sub-assemblies and more details are found in Riyas and Mohanakrishnan (2008). The decay constants of the neutron precursors and the delayed neutron fraction for the PFBR are given as  $\lambda_1 = 0.0129 s^{-1}$ ,  $\lambda_2 = 3.12 s^{-1}$ ,  $\lambda_3 = 0.1344 s^{-1}$ ,  $\lambda_4 = 0.3448 s^{-1}$ ,  $\lambda_5 = 1.3922 s^{-1}$ ,  $\lambda_6 = 3.7491 s^{-1}$ ,  $\beta_1 = 0.00008246$ ,  $\beta_2 = 0.00076817$ ,  $\beta_3 = 0.00066296$ ,  $\beta_4 = 0.0012849$ ,  $\beta_5 = 0.00057615$ ,  $\beta_6 = 0.00017213$ , and  $\Lambda = 4.1 \times 10^{-7} s$ . A step reactivity of 50 pcm is inserted in the PFBR. Assuming that the maximum required order of ETD coefficient is 5, ( $S = 6$  in Eqs. (26)), in this case, the upper limit for the time-step (Eq. (36)) is found to be  $h = 3.5 \times 10^{-4}$ . By choosing  $h = 0.0001$ , the power transient is computed using the modified ETD method and the Cohen's method (1958) for an initial period of 100s. The results are given

**Table 1**

The power transient of thermal reactor computed by the modified ETD method and the TSM method ( $\rho_0 = -1\%$ ). The relative errors are shown in brackets.

| Time (s) | Exact values (Nahla, 2011) | Modified ETD ( $S = 17$ ) $h = 0.1$ s | TSM (Nahla, 2011) $h = 0.1$ s |
|----------|----------------------------|---------------------------------------|-------------------------------|
| 0.0      | 1.0000000                  | 1.0000000                             | 1.0000000                     |
| 0.1      | 0.5205643                  | 0.5205644 (1.0E–7)                    | 0.5206784 (2.2E–4)            |
| 0.2      | 0.4880113                  | 0.4880112 (1.0E–7)                    | 0.488022 (2.25E–5)            |
| 0.4      | 0.4702262                  | 0.4702261 (1.0E–7)                    | 0.470226 (2.10E–7)            |
| 0.6      | 0.4562794                  | 0.4562794 (0.0)                       | 0.4562793 (2.2E–7)            |
| 0.8      | 0.4441506                  | 0.4441506 (0.0)                       | 0.4441505 (2.3E–7)            |
| 1.0      | 0.4333335                  | 0.4333334 (2.3E–7)                    | 0.4333334 (2.3E–7)            |
| 10.0     | 0.2361107                  | 0.2361107 (0.0)                       | 0.2361107 (0.0)               |

**Table 2**

The power transient of thermal reactor computed by the modified ETD method and other methods ( $\rho_0 = 0.003$ ). The relative errors are shown in brackets.

| Time (s) | Exact values (Li et al., 2009) | Modified ETD ( $S = 7$ ) $h = 0.01$ s | TPM (Li et al., 2009) | BBF (Li et al., 2009) $h = 0.1$ s | EFM (Li et al., 2009) |
|----------|--------------------------------|---------------------------------------|-----------------------|-----------------------------------|-----------------------|
| 0.0      | 1.00000                        | 1.00000                               | 1.00000               | 1.00000                           | 1.0000                |
| 0.2      | 1.851268                       | 1.851268 (0.0)                        | 1.851246 (1.2E–6)     | 1.851250 (9.7E–6)                 | 1.851252 (8.6E–6)     |
| 0.4      | 1.947593                       | 1.947593 (0.0)                        | 1.947581 (6.2E–6)     | 1.947585 (4.1E–6)                 | 1.947586 (3.6E–6)     |
| 0.6      | 2.037922                       | 2.037922 (0.0)                        | 2.037915 (3.4E–6)     | 2.037917 (2.5E–6)                 | 2.037918 (2.0E–6)     |
| 0.8      | 2.124832                       | 2.124832 (0.0)                        | 2.124827 (2.4E–6)     | 2.124829 (1.4E–6)                 | 2.124829 (1.4E–6)     |
| 1.0      | 2.209841                       | 2.209840 (4.5E–7)                     | 2.209838 (1.4E–6)     | 2.209839 (9.1E–7)                 | 2.209839 (9.1E–7)     |

**Table 3**

The power transient of the Indian Prototype Fast Breeder Reactor computed by the modified ETD method and the Cohen's method (step reactivity = 50 pcm).

| Time (s) | Modified ETD $h = 0.0001$ s | Cohen $h = 0.0001$ s | Relative error |
|----------|-----------------------------|----------------------|----------------|
| 0.0      | 1.000000                    | 1.000000             | 0              |
| 10.0     | 2.451082                    | 2.451120             | 1.55E–05       |
| 20.0     | 4.522973                    | 4.523096             | 2.72E–05       |
| 30.0     | 8.237472                    | 8.237786             | 3.81E–05       |
| 40.0     | 14.926962                   | 14.927707            | 4.99E–05       |
| 50.0     | 26.986656                   | 26.988310            | 6.13E–05       |
| 60.0     | 48.736440                   | 48.739983            | 7.27E–05       |
| 70.0     | 87.969711                   | 87.977112            | 8.41E–05       |
| 80.0     | 158.746774                  | 158.761943           | 9.55E–05       |
| 90.0     | 286.434427                  | 286.465074           | 1.07E–04       |
| 100.0    | 516.798131                  | 516.859342           | 1.18E–04       |

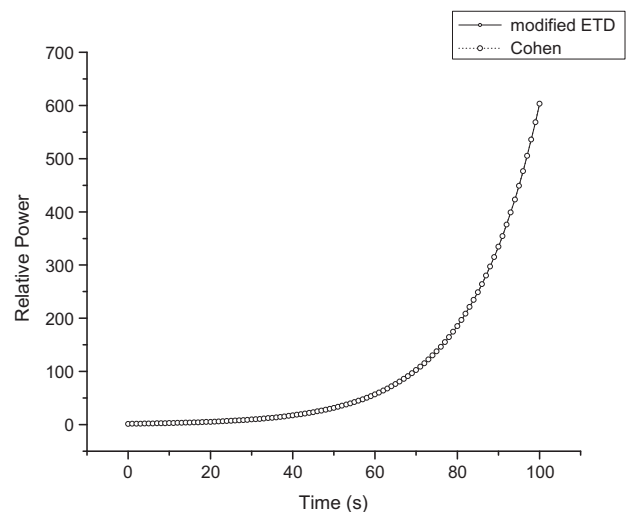
in Table 3. The power transient computed by this method is in good agreement with the Cohen's method (1958). This ensures that the modified ETD method is capable of estimating the power transients accurately for longer duration of time for a typical fast breeder reactor. The power transients obtained using the modified ETD method and the Cohen's method are shown in Fig. 1. A maximum relative error of  $1.18\text{E}-4$  is observed during the transient time.

#### 4.2. Power transients of thermal and fast reactors caused by ramp reactivity

In this section we consider the power transients, caused by positive and negative ramp reactivity, of the thermal reactor described by Li et al. (2009) in Section 4.1 First consider the positive ramp reactivity  $\rho(t) = 0.1\text{ t}\$/\text{s}$ . The power transients, following this ramp reactivity, are estimated using the modified ETD method using a time-step  $h = 0.01\text{ s}$  and  $S = 7$ . The results are compared with the Taylor's series method (TSM), the better basis function method (BBF) and the Hermite Polynomial Method (HPM) referred by Nahla (2011) and Li et al. (2009). The results are shown in Table 4. The relative errors are shown in brackets in Table 4. It is observed from Table 4 that the modified ETD method gives more accurate results with larger time-step as compared to the TSM (Nahla, 2011). In this example the relative error at  $t = 10\text{ s}$  is computed for various orders of ETD coefficients. It is observed that as the order of the ETD coefficients is increased, the relative error becomes smaller and it converges fast. This is shown in Fig. 2.

The power transient of the thermal reactor (Nahla, 2011) caused by the negative ramp reactivity  $\rho(t) = -0.1\text{ t}\$/\text{s}$  is also computed using the modified ETD method and the results are shown in Table 5. Table 5 also shows the power transients estimated by other standard methods (Nahla, 2011). Since the exact values of the power transients are not reported in this case, the relative errors are not given in Table 5. It is observed from Table 5 that the power transients estimated by this method are in good agreement with other methods.

Consider the power transient of the Indian PFBR, described in Section 4.1., caused by a ramp reactivity of  $1\text{ pcm/s}$ . Table 6 shows the power transient computed using the modified ETD method and the Cohen's method (1958) for an initial period of  $100\text{ s}$ . The relative errors are also shown in Table 6. The power transients estimated by this method is in good agreement with the Cohen's method (1958). A maximum relative error of  $1.83\text{E}-04$  is observed during this transient time. From Table 6 it is observed that this method is capable of estimating the power transients for fast reactors with ramp reactivity input for longer duration of time. Fig. 3



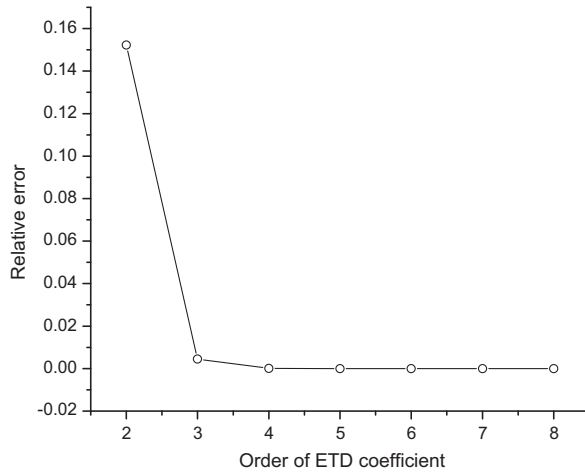
**Fig. 1.** The power transient of the Indian Prototype Fast Breeder Reactor (PFBR) computed by the modified ETD method (solid line) and the Cohen's method (dot) for step reactivity  $\rho(t) = 50\text{ pcm}$ .



**Table 4**

The power transient of thermal reactor computed by the ETD method and other methods for positive ramp reactivity  $\rho_0 = 0.1$  t\$/s. The relative errors are shown in brackets.

| Time (s) | Exact values (Li et al., 2009) | Modified ETD ( $S = 7$ ) $h = 0.01$ s | TSM (Nahla, 2011) $h = 0.0001$ s | BBF (Li et al., 2009) $h = 0.1$ s | HPM (Li et al., 2009) $h = 0.1$ s |
|----------|--------------------------------|---------------------------------------|----------------------------------|-----------------------------------|-----------------------------------|
| 0.0      | 1.000000                       | 1.000000                              | 1.0000                           | 1.00000                           | 1.00000                           |
| 2.0      | 1.338200                       | 1.338200 (0.0)                        | 1.3382 (0)                       | 1.338200 (0.0)                    | 1.33824 (2.98E–5)                 |
| 4.0      | 2.228442                       | 2.228442 (0.0)                        | 2.2284 (1.88E–5)                 | 2.228442 (0.0)                    | 2.22844 (8.97E–7)                 |
| 6.0      | 5.582052                       | 5.582052 (0.0)                        | 5.5822 (2.65E–5)                 | 5.582052 (0.0)                    | 5.58204 (2.14E–6)                 |
| 8.0      | 42.78630                       | 42.78630 (0.0)                        | 42.789 (6.31E–5)                 | 42.78629 (2.33E–7)                | 42.7862 (2.33E–6)                 |
| 10.0     | 451163.6                       | 451163.6 (0.0)                        | 451430 (5.9E–4)                  | 450412.1 (1.66E–3)                | 447385.5 (8.37E–3)                |



**Fig. 2.** The variation of relative error with the order of the ETD coefficient while estimating the power transients of thermal reactor, described by Li et al. (2009), at  $t = 10$  s, for a positive ramp reactivity  $\rho(t) = 0.1$  t\$/s.

shows the power transients computed using the modified ETD method and the Cohen's method.

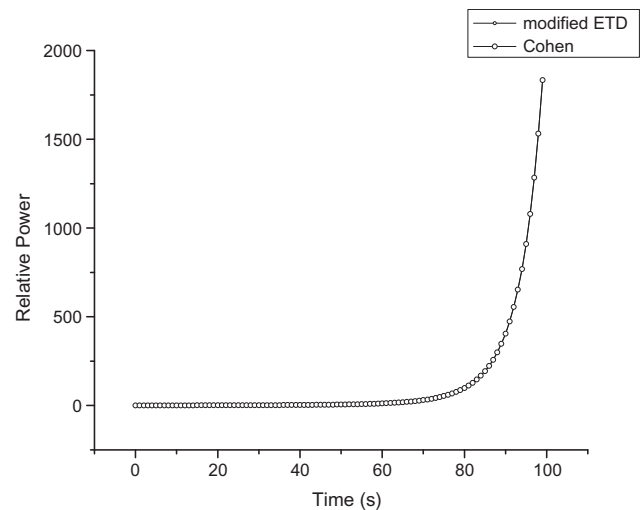
#### 4.3. Power transients of thermal reactor caused by oscillatory reactivity and high reactivity insertion rate

The power transients caused by a sinusoidal reactivity insertion is analysed here for the thermal reactor described by Li et al. (2009) in Section 4.1. Consider a sinusoidal reactivity of the form  $\rho(t) = 0.001 \sin(4\pi t)$ . The power transients following this sinusoidal reactivity are estimated using the modified ETD method ( $h = 0.001$  s,  $S = 7$ ) and compared with the TSM (Nahla, 2011) method. The results are shown in Table 7. Consider another severe accidental scenario where there is a large reactivity insertion rate, i.e.,  $\rho(t) = 0.5$  t\$/s for  $0 \leq t \leq 1$  s and  $\rho(t) = 0.5$  for  $1 \leq t \leq 4$  s. This kind of large reactivity insertion rate may be observed during the time of criticality when there is an uncontrolled withdrawal of control rod. The power transients caused by this kind of large reactivity insertion rate is also analysed for the thermal reactor (Li et al., 2009) using the modified ETD method ( $h = 0.01$  s,  $S = 7$ ) and the results are compared with the TSM (Nahla, 2011) method. The results are given in Table 8. From these results it is assured

**Table 6**

The power transient of the Indian Prototype Fast Breeder Reactor (PFBR) computed by the modified ETD method and the Cohen's method for positive ramp reactivity of 1 pcm/s.

| Time (s) | Modified ETD ( $S = 3$ ) $h = 0.0001$ s | Cohen $h = 0.0001$ s | Relative Error |
|----------|-----------------------------------------|----------------------|----------------|
| 0.0      | 1.000000                                | 1.000000             | 0.0            |
| 10.0     | 1.102048                                | 1.102051             | 2.72E–06       |
| 20.0     | 1.353823                                | 1.353832             | 6.65E–06       |
| 30.0     | 1.860535                                | 1.860560             | 1.34E–05       |
| 40.0     | 2.895502                                | 2.895566             | 2.21E–05       |
| 50.0     | 5.189995                                | 5.190173             | 3.43E–05       |
| 60.0     | 10.939190                               | 10.939743            | 5.05E–05       |
| 70.0     | 27.769447                               | 27.771452            | 7.22E–05       |
| 80.0     | 87.199901                               | 87.208655            | 1.00E–04       |
| 90.0     | 348.924903                              | 348.972575           | 1.37E–04       |
| 100.0    | 1839.603459                             | 1839.939691          | 1.83E–04       |



**Fig. 3.** The power transient of the Indian Prototype Fast Breeder Reactor (PFBR) computed by the modified ETD method (solid line) and the Cohen's method (dot) for positive ramp reactivity of 1 pcm/s.

that the modified ETD method is capable of estimating the power transients in a more realistic way for oscillatory reactivity insertion and high reactivity insertion rate also.

**Table 5**

The power transient of thermal reactor computed by the modified ETD method and other methods for negative ramp reactivity  $\rho_0 = -0.1$  t\$/s.

| Time (s) | Modified ETD ( $S = 17$ ) $h = 0.01$ s | TSM (Nahla, 2011) $h = 0.001$ s | GAEM (Nahla, 2011) $h = 0.001$ s | Pade (Nahla, 2011) $h = 0.001$ s |
|----------|----------------------------------------|---------------------------------|----------------------------------|----------------------------------|
| 0.0      | 1.000000                               | 1.0000                          | 1.000000                         | 1.00000                          |
| 2.0      | 0.792005                               | 0.791955                        | 0.792007                         | 0.792007                         |
| 4.0      | 0.613016                               | 0.612976                        | 0.613020                         | 0.613018                         |
| 6.0      | 0.474057                               | 0.474027                        | 0.474065                         | 0.474058                         |
| 8.0      | 0.369168                               | 0.369145                        | 0.369172                         | 0.369169                         |
| 10.0     | 0.290654                               | 0.290636                        | 0.290653                         | 0.290654                         |

**Table 7**

The power transient of thermal reactor computed by the modified ETD method and the TSM method for sinusoidal reactivity  $\rho = 0.001 \sin(4\pi t)$ ,  $0 \leq t \leq 5$  s).

| Time (s) | $\rho$ (t)  | Modified ETD ( $S = 7$ ) $h = 0.001$ s | TSM $h = 0.001$ s |
|----------|-------------|----------------------------------------|-------------------|
| 0        | 0           | 1.0000000                              | 1.0000000         |
| 0.4      | -9.51102E-4 | 0.8828488                              | 0.8828564         |
| 0.8      | -5.87545E-4 | 0.9334387                              | 0.9333666         |
| 1.2      | 5.88145E-4  | 1.1070115                              | 1.1069329         |
| 1.6      | 9.50873E-4  | 1.1615297                              | 1.1615597         |
| 2.0      | -7.41229E-7 | 0.9992939                              | 0.9993774         |
| 2.4      | -9.51331E-4 | 0.8857330                              | 0.8857506         |
| 2.8      | -5.86945E-4 | 0.9366979                              | 0.9366357         |
| 3.2      | 5.88744E-4  | 1.1108737                              | 1.1107949         |
| 3.6      | 9.50643E-4  | 1.1653616                              | 1.1654019         |
| 4.0      | -1.48246E-6 | 1.0024811                              | 1.0025649         |
| 4.4      | -9.51559E-4 | 0.8885748                              | 0.8885925         |
| 4.8      | -5.86345E-4 | 0.9397346                              | 0.9396721         |
| 5.0      | -1.85307E-6 | 1.0039612                              | 1.0040551         |

**Table 8**

The power transient of thermal reactor computed by the modified ETD method and the TSM method for a mixture of positive ramp reactivity  $\rho = 0.5$  t/s ( $0 \leq t \leq 1$  s) and step reactivity  $\rho_0 = 0.5$  ( $1 \leq t \leq 4$  s).

| Time (s) | $\rho(t)$ | Modified ETD ( $S = 7$ ) $h = 0.01$ s | TSM $h = 0.001$ s |
|----------|-----------|---------------------------------------|-------------------|
| 0        | 0         | 1.0000000                             | 1.0000000         |
| 0.2      | 0.1       | 1.1137105                             | 1.1137097         |
| 0.4      | 0.2       | 1.2688773                             | 1.2688772         |
| 0.6      | 0.3       | 1.4836057                             | 1.4836057         |
| 0.8      | 0.4       | 1.7932624                             | 1.7932623         |
| 1.0      | 0.5       | 2.2671657                             | 2.2671655         |
| 1.2      | 0.5       | 2.4287809                             | 2.4287808         |
| 1.4      | 0.5       | 2.5730775                             | 2.5730773         |
| 1.6      | 0.5       | 2.7157281                             | 2.7157279         |
| 1.8      | 0.5       | 2.8586116                             | 2.8586114         |
| 2.0      | 0.5       | 3.0029664                             | 3.0029662         |
| 2.5      | 0.5       | 3.3756185                             | 3.3756183         |
| 3.0      | 0.5       | 3.7720567                             | 3.7720565         |
| 3.5      | 0.5       | 4.1987588                             | 4.1987586         |
| 4.0      | 0.5       | 4.6609933                             | 4.6609930         |
| 4.5      | 0.5       | 5.1635628                             | 5.1635626         |
| 5.0      | 0.5       | 5.7111850                             | 5.7111847         |

## 5. Conclusions

The modified exponential time differencing (ETD) method is developed for solving the point kinetics equations with multi group of delayed neutrons. The advantage of using the modified ETD method is that the power transient may be computed using large time-step for various types of reactivity perturbations. This

method is applied to estimate the power transients of both thermal reactor and the Indian Prototype Fast Breeder Reactor (PFBR) with multi group of delayed neutrons. Three types of reactivity perturbations, i.e., step, ramp and oscillatory input are discussed and the results are compared with other standard methods. From the comparison of results, it is observed that the modified exponential time differencing method estimates the power transients accurately for both thermal and fast reactors with multi group of delayed neutrons. This method is also tested for estimating the power transients with multi group of delayed neutrons in a more realistic way using a high reactivity insertion rate. The results are found to be in good agreement with other methods. This ensures that this method is capable of estimating the power transients under severe accidental scenario also. A criterion to choose the time-step and its upper bound are also derived and presented. It is also shown that as the order of ETD coefficients is increased, the global errors involved in the computation of power transients and precursor concentrations become smaller. This method is a self-starting method and does not require any previous power history.

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